Measuring Strategic Hedging in Latin America: A Multilayer Network Approach

PhD research proposal

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Research Problem

The intensification of great power rivalry between the United States, China, and Russia has placed weaker states under mounting pressure to align with one pole or another, making the study of alignment strategies one of the most pressing questions in contemporary international relations. This research project aims to classify the alignment behaviour of 20 Latin American states through measurable indicators, using multilayer network analysis.

This work builds upon Sims and Lee's (2025) Hedging Index for Latin America, which blends bilateral trade shares with arms procurement data from SIPRI's Trend Indicator Value series. While methodologically innovative, the index conflates three analytically distinct phenomena, namely structural legacy dependency, reactive adaptation to external actor behaviour, and genuine strategic hedging. The Nicaragua case illustrates this most starkly. A government openly hostile to Washington scores 0.946 on the US-alignment scale, almost identical to Honduras, which hosts a long-standing US military presence at Soto Cano Air Base. This result reflects not strategic alignment but the invisibility of Russia, Nicaragua's actual arms supplier, within a strictly bilateral US-China measurement framework. More broadly, the index treats trade flows and arms sales as signals of political alignment when they are often better understood as products of market structure and supplier dominance.

To address these limitations, this research proposes to measure alignment behaviour across Latin America using multilayer network analysis, drawing on a richer set of indicators to map the evolving structure of great power competition in the hemisphere. Scholars have long examined how secondary states navigate the pressure resulting from great power rivalry through three primary strategies. Balancing involves forming coalitions to counter the strongest power (Walt, 1987; Waltz, 1979), bandwagoning involves aligning with a dominant or threatening power for protection or gain (Schweller, 1994), and hedging involves engaging multiple great powers simultaneously while avoiding firm commitment to any single actor (Goh, 2005; Haacke, 2019; Kuik, 2021). The hedging literature has advanced considerably since Medeiros (2005) lamented that the

concept was "highly underdeveloped both in international relations theory and security studies literature", but it remains heavily focused on Southeast Asia.

Koga (2018) revisited the concept systematically, situating it within the balancing-bandwagoning spectrum and arguing that states adopt hedging as a third strategic option under conditions of power shift and strategic uncertainty. Crucially, Koga argues that economic and military indicators alone are insufficient to classify alignment behaviour and that diplomatic factors must be incorporated as a complementary, if weaker, signal. Kuik (2021) developed this further, arguing that hedging is best understood not as a calculated grand strategy but as instinctive insurance-seeking behaviour driven by uncertainty about both the principal threat and the reliability of potential patrons, and that the concept travels beyond Southeast Asia to other regions facing analogous structural conditions. Despite this theoretical progress, systematic empirical measurement of alignment behaviour remains underdeveloped, particularly outside East Asia. Latin America is emerging as a central arena of geopolitical and geoeconomic competition, where the United States seeks to "reassert and enforce the Monroe Doctrine to restore American preeminence (…) [and] deny non-Hemispheric competitors the ability to position forces or other threatening capabilities" (The White House, 2025), yet the region remains understudied in the quantitative hedging literature.

The objective of this research is to address precisely this gap by investigating how Latin American states navigate great-power competition through multilayer network analysis. Specifically, it addresses three persistent limitations in existing measurement approaches.

The first limitation concerns the conflation of pre-existing structural dependency with voluntary political alignment. Existing approaches that use bilateral trade flows and arms procurement data as alignment proxies risk measuring market structure rather than strategic preference. China is the dominant trading partner for the majority of Latin American states regardless of their political orientation, reflecting global commodity demand rather than deliberate alignment choice. US arms exports reflect supplier dominance and interoperability ecosystems rather than bilateral political deference. A country purchasing ammunition from the US or exporting copper to Chinese refineries is not necessarily signalling political alignment in either direction.

The second limitation concerns the conflation of reactive adaptation with strategic hedging. As Koga (2018) notes, identifying hedging requires distinguishing it from mere "waiting for balancing", and Kuik (2021) emphasises that states hedge because structural conditions compel them to, not because they have designed a deliberate multi-vector strategy. The empirical record from Latin America illustrates this distinction sharply. Bolivia's rapid reorientation toward Washington following the 2025 election reflects reactive adaptation to domestic political change rather than a sustained hedging strategy, while Peru has joined Trump's Shield of the Americas and continues to cooperate with China, with which it jointly operates the Chancay mega-port north of Lima. Collapsing these cases into a single alignment continuum obscures analytically important variation.

The third limitation is the exclusive focus on the US-China dyad, which excludes other strategically significant actors. Russia's security ties with Cuba, Nicaragua and, until the US intervention,

Venezuela, and the EU's normative and investment engagement through the Global Gateway across the region all constitute alignment-relevant relationships that a strictly bilateral framework cannot capture.

Theoretical Contribution

The central theoretical contribution is a distinction between three forms of alignment behaviour that existing frameworks collapse into a single continuum. Pre-existing structural dependency refers to inherited trade, financial, and security ties that constrain short-term choices regardless of political preference, such as Argentina's reliance on China's currency swap line or Brazil's dependence on Chinese soybean markets, which will likely persist regardless of who governs Brazil in the coming years. Reactive adaptation describes alignment shifts driven by external actor behaviour rather than deliberate strategic design. Examples include Bolivia's rapid reorientation toward Washington under Paz following decades of estrangement under Morales, Honduras' alignment with Washington following Trump's explicit electoral intervention and the pardon of former president Hernández, who had been sentenced to 45 years in prison by a US court for drug trafficking, and Colombia's foreign-policy shift under Abelardo de la Espriella after Gustavo Petro's mandate. Strategic hedging, by contrast, denotes deliberate, sustained management of relationships with competing great powers to preserve policy autonomy, as illustrated by Peru's purchase of US F-16 fighter jets alongside its acceptance of the Chinese-financed and jointly operated Chancay mega-port.

Recent developments in Latin America provide ample empirical evidence for this theoretical debate. In multiple cases, what appears as multi-vector engagement is better explained by the simultaneous pull of Chinese economic indispensability and US political coercion, a dynamic that intensified markedly following the 2025 US National Security Strategy's explicit invocation of a Trump Corollary to the Monroe Doctrine.

Measurement Framework

The proposed Multilayer Hedging Index replaces the Sims and Lee bilateral trade and arms dyad with a set of indicators selected specifically to capture voluntary political alignment rather than structural dependency. The indicators are grouped into three layers, each capturing a distinct dimension of alignment behaviour.

The first layer covers diplomatic and institutional signalling. It includes UNGA voting patterns on contested resolutions, such as votes on Ukraine, Gaza, and human rights mechanisms; membership of the Americas Counter-Cartel Coalition (ACCC, also known as Trump's Shield of the Americas), which currently comprises 19 member nations and serves as an explicit US alignment instrument; membership of the World AI Cooperation Organisation (WAICO), which Beijing launched in 2026 in competition with the US-anchored Pax Silica framework; continued membership of, or withdrawal from, the International Criminal Court, given US pressure to exit the institution; and free trade agreements with China, which Beijing has concluded with Chile, Costa Rica, Ecuador, Nicaragua, and Peru.

The second layer covers economic and financial alignment signals that go beyond raw trade volumes. It includes the stock of Chinese foreign direct investment relative to GDP, which captures

structural economic penetration more meaningfully than bilateral trade flows; membership of the Belt and Road Initiative (BRI) and the status of specific BRI projects; Huawei's penetration of national 5G infrastructure; and the imposition of US tariffs or equivalent measures, as a proxy for Washington's assessment of each country's alignment.

The third layer covers security alignment signals. It includes membership of the Shield of the Americas; engagement with US-linked private security actors, including cooperation agreements involving entities associated with Erik Prince, the founder of Blackwater, in Ecuador and Haiti; bilateral security agreements with the US; and the nature and conditionality of military assistance.

Expected Contribution

This research makes three distinct contributions. Theoretically, it aims to provide a systematic operationalisation of the distinction between reactive adaptation and strategic hedging in Latin America. Methodologically, it introduces a multilayer network approach to alignment measurement that is more sensitive to the multidimensional nature of great power competition than existing single-index approaches. Empirically, it produces a comprehensive, reproducible alignment dataset for Latin America at a moment when the region is undergoing its most significant geopolitical reorientation since the Cold War.

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